LoRMA neural retriever: Methodology and Evidence

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Status. Post-run methodology reconstruction grounded in archived source and reports. This document is not an executed historical notebook.

Research question

Try parameter-efficient retrieval adaptation instead of updating the whole Phase 1 model. The GPU report lists about 1.18 million trainable parameters for this branch.

Architecture and mechanism

LoRMA low-rank modules alter the inherited A0 representation. The query/candidate window encoder and late interaction stay comparable to the full branch.

Measured result

Scored submission	1039891
Codabench micro-F1@7	0.3969
Archived local metric	0.3969 corrected local-dev micro-F1@7

Corrected local dev micro-F1@7 was 0.3969 and Codabench was also 0.3969 to four decimals. The result did not beat the full fine-tune here.

Limitations and next check

Parameter count and score alone do not measure training cost or memory; the run log is needed for those comparisons. Local, pilot, and hidden metrics are deliberately kept distinct. A single scored run does not establish a causal architecture effect or statistical significance.

Source and reproducibility

Original source modules: phase7_run.py, phase7_export.py. SHA-256 hashes, the original report, and the prediction ZIP receipt are in the companion provenance file. The explanatory notebook was written after the historical run; original execution output is never implied. Reproduction requires authorized course data, the Phase 1 parent weights/tokenizer, and the dependencies in the archived source.

Documentation disclosure

AI assistance was used to organize the archived evidence and draft this explanatory document. All numerical claims above come from saved run records or the provided finished-submission screenshot; missing evidence is labelled as such.